Category Theory

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March 2025

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In classical set theory, cutting a set or a space is the standard way to study it. We would like to ask what elements are in the space.
However, in category theory, objects in a category is some kind of black box without inner structure. The classical way fails to study an object because we are not allowed to cut it into pieces. So, how would we study an object without penetrating it?
The philosophy that an object is fully determined by its relations between it and any other objects in a universe helps us. This philosophy is perfectly reflected by the following theorem.

Lemma 2.1: Hom functor

Let $\mathscr{C}$ be a category. There is a covariant functor

$$G : \mathscr{C} \to \mathbf{Set}^{\mathscr{C}^{\mathrm{op}}}$$

sending objects X to $\mathrm{Hom}_{\mathscr{C}}(-, X)$ and morphisms $f : X \to Y$ to $G(f) : \{\mathrm{Hom}_{\mathscr{C}}(A, X) \to \mathrm{Hom}_{\mathscr{C}}(A, Y)\}_A$ where $G(f)$ is the natural transformation: $\forall$ morphism in category $\mathscr{C}$, $u : A \to B$, the following diagram commutes

$$\begin{tikzcd}
\mathrm{Hom}_{\mathscr{C}}(A, X) \arrow[r, "G(f)_A"] & \mathrm{Hom}_{\mathscr{C}}(A, Y) \\
\mathrm{Hom}_{\mathscr{C}}(B, X) \arrow[r, "G(f)_B"] \arrow[u, "\mathrm{Hom}_{\mathscr{C}}(-,X)(u)"] & \mathrm{Hom}_{\mathscr{C}}(B, Y) \arrow[u, "\mathrm{Hom}_{\mathscr{C}}(-,Y)(u)"']
\end{tikzcd}$$

Remark This functor is not nothing or unvaluable. In fact, it is invaluable because it implies a shift of paradigm. The left hand side is the category $\mathscr{C}$ that we are interested, but we might know little about its objects X. Although the category on the right hand side $\mathbf{Set}^{\mathscr{C}^{\mathrm{op}}}$ looks not understandable. But, its objects are contravariant functors $\mathscr{C} \to \mathbf{Set}$. The functor G shifts our attention from an object X that we might be unfamiliar or is not able to dissect, to an object $\mathrm{Hom}_{\mathscr{C}}(-, X)$ that is all morphisms between X and every object in $\mathscr{C}$. This is a relative point of view (*point du vue relatif*): Instead of looking at the interior part(=elements) of an object X, we can know structures of that X by looking at how it interacts with other objects(=morphisms between it and other objects) in the same universe(= $\mathrm{Ob}(\mathscr{C})$).

Lemma 2.2: Yoneda Lemma(First encounter)

Let $\mathscr{C}$ be a category and $X \in \mathrm{Ob}(\mathscr{C})$ be an *arbitrary* object. Then, consider the contravariant functor $h_X := \mathrm{Hom}_{\mathscr{C}}(-, X)$ that $h_X : \mathscr{C} \to \mathbf{Set}$. $\forall$ contravariant functor $F : \mathscr{C} \to \mathbf{Set}$, $\exists$ a bijection between the set of natural transformations $h_X \rightsquigarrow F$ and $F(X)$.

Remark Since both h_X and F are contravariant functors from $\mathscr{C} \to \mathbf{Set}$, $h_X, F \in \mathrm{Ob}(\mathbf{Set}^{\mathscr{C}^{\mathrm{op}}})$ by the definition of functor category and the lemma 2.1. The morphisms in this functor category $\mathbf{Set}^{\mathscr{C}^{\mathrm{op}}}$ are natural transformations. So, this lemma can be packed in a more abstract language,

$$\boxed{\forall X \in \mathrm{Ob}(\mathscr{C}), \forall F \in \mathrm{Ob}(\mathbf{Set}^{\mathscr{C}^{\mathrm{op}}}), \exists \text{ a bijection } \mathrm{Hom}_{\mathbf{Set}^{\mathscr{C}^{\mathrm{op}}}}(h_X, F) \longleftrightarrow F(X)}$$

This is parallel to the lemma 2.1 in the following sense. If the previous lemma tells us to shift the prespective when studying an object, this Yoneda lemma is a functor version mimicing the lemma 2.1. This package $\mathrm{Hom}_{\mathbf{Set}^{\mathscr{C}^{\mathrm{op}}}}(h_X, F)$ is not an abstract nonsense. Instead, it tells us the philosophy that to study an alien functor F, it suffices to think about how it interacts with the 'standard' functor h_X. This 'interaction' is packed in the symbol $\mathrm{Hom}_{\mathbf{Set}^{\mathscr{C}^{\mathrm{op}}}}(h_X, F)$.

Proof: Let's first construct the map $\Phi : \mathrm{Hom}_{\mathbf{Set}^{\mathscr{C}^{\mathrm{op}}}}(h_X, F) \to F(X)$ and $\Psi : F(X) \to \mathrm{Hom}_{\mathbf{Set}^{\mathscr{C}^{\mathrm{op}}}}(h_X, F)$. Then, we want to show that $\Phi \circ \Psi = \mathrm{id}_{F(X)}$ and $\Psi \circ \Phi = \mathrm{id}_{\mathrm{Hom}_{\mathbf{Set}^{\mathscr{C}^{\mathrm{op}}}}(h_X, F)}$.
Define $\Phi : \mathrm{Hom}_{\mathbf{Set}^{\mathscr{C}^{\mathrm{op}}}}(h_X, F) \to F(X)$ to be

$$\Phi : \alpha \mapsto \alpha_X(\mathrm{id}_X)$$ [1]

and Ψ to be

$$\Psi : f \mapsto \beta, \quad \beta_A : \varphi \mapsto F(\varphi)(f)$$ [2]

First, $\forall f \in F(X)$,

$$\begin{aligned}
\Phi(\Psi(f)) &= \Phi(\beta) && \text{Set } \beta := \Psi(f) \\
&= \beta_X(\mathrm{id}_X) && \text{definition of } \Phi \\
&= F(\mathrm{id}_X)(f) && \text{definition of } \beta_X \\
&= \mathrm{id}_{F(X)}(f) = f && F \text{ is a functor}
\end{aligned}$$

To see that whether $\Psi(\Phi(\alpha)) = \alpha$ for any natural transformation α. We need to check:
(1) Whether $\Psi(\Phi(\alpha))$ is a natural transformation.
(2) $\forall A \in \mathrm{Ob}(\mathscr{C})$, $\Psi(\Phi(\alpha))_A = \alpha_A$.
For (1), let $\gamma := \Psi(\Phi(\alpha))$. Applying $\gamma = \Psi(\Phi(\alpha))$ to an arbitrary morphism $u : A \to B$, we have

$$\begin{tikzcd}
h_X(A) \arrow[r, "\gamma_A"] & F(A) \\
h_X(B) \arrow[u, "h_X(u)"] \arrow[r, "\gamma_B"] & F(B) \arrow[u, "F(u)"']
\end{tikzcd}$$

For any morphism $\varphi : B \to X$,

$$\begin{aligned}
(\gamma_A \circ h_X(u))(\varphi) &= \gamma_A(\varphi \circ u) \\
&= \Psi(\alpha_X(\mathrm{id}_X))_A(\varphi \circ u) \\
&= F(\varphi \circ u)(\alpha_X(\mathrm{id}_X)) \\
&= F(u) \circ F(\varphi)(\alpha_X(\mathrm{id}_X))
\end{aligned}$$

While, $(F(u) \circ \gamma_B)(\varphi) = F(u)\big(F(\varphi)(\alpha_X(\mathrm{id}_X))\big)$, which implies that this diagram is commutative.
For (2), for an arbitrary morphism $\eta : A \to X$. There is a commutative diagram

$$\begin{tikzcd}
h_X(X) \arrow[r, "\alpha_X"] \arrow[d, "h_X(\eta)"'] & F(X) \arrow[d, "F(\eta)"] \\
h_X(A) \arrow[r, "\alpha_A"] & F(A)
\end{tikzcd}$$

Then, applying γ_A to this η, we have

$$\begin{aligned}
\Psi(\Phi(\alpha))_A(\eta) &= \Psi(\alpha_X(\mathrm{id}_X))_A(\eta) \\
&= F(\eta)(\alpha_X(\mathrm{id}_X)) \\
&= \alpha_A(h_X(\eta)(\mathrm{id}_X)) && \text{by the commutativity of the diagram} \\
&= \alpha_A(\eta) && h_X(\eta)(\mathrm{id}_X) = \mathrm{id}_X \circ \eta = \eta
\end{aligned}$$

Thereby, we conclude that $\Psi(\Phi(\alpha)) = \alpha$. □

[1] This is because $\forall$ A$\in$ Ob($\mathscr{C}$, α_A is a morphism $\mathrm{Hom}_{\mathscr{C}}(A, X) \to F(A)$. If we want something in $F(X)$, set $A = X$. That's why we get

$$\alpha_X : \mathrm{Hom}_{\mathscr{C}}(X, X) \to F(X)$$

We just need one input in $\mathrm{Hom}_{\mathscr{C}}(X, X)$ so then we have a desired output in $F(X)$. But what kind of input exist and is independent of category $\mathscr{X}$, object X? That must be the identity map id_X. That is how we set the right hand side $\alpha_X(\mathrm{id}_X)$.

[2] β is a natural transformation, so it has to be evaluated on arbitrary object $A \in \mathrm{Ob}(\mathscr{C})$. On each A, how is a morphism $\varphi \in h_X(A) = \mathrm{Hom}_{\mathscr{C}}(A, X)$ being sent to $F(A)$? Notice that F is contravariant, so $F(\varphi) : F(X) \to F(A)$. f is an object in $F(X)$, so it is a possible input of $F(\varphi)$.

Corollary 2.1: Rrepresentable Yoneda

3 Adjunctions

Theorem 3.1: Adjoint functors and limits

Let $(F \dashv G)$ be an adjoint pair of covariant functors such that $F : \mathscr{C} \to \mathscr{D}$ and $G : \mathscr{D} \to \mathscr{C}$. Then, F preserves all direct limits and G preserves all inverse limit.

Proof: Let $(\{C_i\}, \{f_{ij}\})$ be a direct system indexed by I. Let $\varinjlim C_i$ be the direct limit in $\mathscr{C}$. So, we have already had the following commutative diagram:

$$\begin{tikzcd}
C_i \arrow[rr, "f_{ij}"] \arrow[dr, "\alpha_i"'] \arrow[ddr, bend right, "\beta_i"'] & & C_j \arrow[dl, "\alpha_j"] \arrow[ddl, bend left, "\beta_j"] \\
& \varinjlim C_i \arrow[d, "\exists ! \Phi"'] & \\
& \widetilde{C} &
\end{tikzcd}$$

Figure 1: Commutative diagram for the direct limit

Applying a covariant functor F to the diagram 1 , we have:

$$\begin{tikzcd}
F(C_i) \arrow[rr, "Ff_{ij}"] \arrow[dr, "F\alpha_i"'] & & F(C_j) \arrow[dl, "F\alpha_j"] \\
& F(\varinjlim C_i) &
\end{tikzcd}$$

(a) (partial diagram)

$$\begin{tikzcd}
F(C_i) \arrow[rr, "Ff_{ij}"] \arrow[dr, "F\alpha_i"'] \arrow[ddr, bend right, "\gamma_i"'] & & F(C_j) \arrow[dl, "F\alpha_j"] \arrow[ddl, bend left, "\gamma_j"] \\
& F(\varinjlim C_i) \arrow[d, dashed, "\exists ! h"'] & \\
& D &
\end{tikzcd}$$

(b) (whole diagram)

Figure 2: Commutative diagram for the direct limit being apllied by F

By covariant property, since $\alpha_i = \alpha_j \circ f_{ij}$, then $F(\alpha_i) = F(\alpha_j \circ f_{ij}) = F(\alpha_j) \circ F(f_{ij})$, so this diagram commutes.

We would like to show that $F(\varinjlim C_i) \cong \varinjlim F(C_i)$ (since it is unique up to isomorphism) i.e. $\exists !\ h : F(\varinjlim C_i) \to D$ such that $\forall D \in \mathrm{Ob}(D)$, and $\forall$ morphisms $\gamma_i : F(C_i) \to D$, $\gamma_i = h \circ F\alpha_i$, i.e. making the following diagram commute: (just an expansion of diagram 2) So it is split into two parts: (1.1) Existence of such h that commutes with those morphisms (1.2) Uniqueness

Even though $GF \neq \mathbb{1}_{\mathscr{C}}$, we can line up this with C_i s: Since F as a left adjoint functor gives unit, $\eta : \mathbb{1}_{\mathscr{C}} \to GF$, which is a natural equivalence (natural isomorphism). So, it gives

- a collection of isomorphisms $\eta_{C_i} : C_i \xrightarrow{\sim} GF(C_i) \in \mathrm{Mor}_{\mathscr{C}}(C_i, GFC_i)$

$$
\begin{tikzcd}
GF(C_i) \arrow[rr, "GFf_{ij}"] \arrow[dr, "GF\alpha_i"'] \arrow[ddr, bend right, "G\gamma_i"'] & & GF(C_j) \arrow[dl, "GF\alpha_j"] \arrow[ddl, bend left, "G\gamma_j"] \\
& GF(\varinjlim C_i) \arrow[d, dashed, "?"] & \\
& GD &
\end{tikzcd}
$$

Figure 3

$$
\begin{tikzcd}
& & C_j \arrow[rrr, dashed, "\eta_{C_j}"] \arrow[dl] & & & GFC_j \arrow[dl] \arrow[ddl, bend left, "G\gamma_j"] \\
C_i \arrow[urr, "f_{ij}"] \arrow[rrr, dashed, "\eta_{C_i}"] \arrow[dr] & & & GFC_i \arrow[urr, "GFf_{ij}"] \arrow[dr] \arrow[ddr, bend right, "G\gamma_i"'] & & \\
& \varinjlim C_i \arrow[rrr, dashed, "\eta_{\varinjlim C_i}"] & & & GF(\varinjlim C_i) \arrow[d] & \\
& & & & GD &
\end{tikzcd}
$$

Figure 4

- $\eta_{\varinjlim C_i} : \varinjlim C_i \to GF(\varinjlim C_i)$

by composing η_{C_i} with $G\gamma_i$, we have

$$
\begin{tikzcd}
C_i \arrow[rr, "f_{ij}"] \arrow[dr, "\alpha_i"'] \arrow[ddr, bend right, "G\gamma_i \circ \eta_{C_i}"'] & & C_j \arrow[dl, "\alpha_j"] \arrow[ddl, bend left, "G\gamma_j \circ \eta_{C_j}"] \\
& \varinjlim C_i \arrow[d, "\exists !h'"] & \\
& GD &
\end{tikzcd}
$$

Figure 5

Each chain $G\gamma_i \circ \eta_{C_i}$ is got by composition: $C_i \xrightarrow{\eta_{C_i}} GFC_i \xrightarrow{G\gamma_i} GD$.
By universal property of direct limit, $\exists !$ morphism $h' : \varinjlim C_i \to GD$ such that

$$h' \circ \alpha_i = G\gamma_i \circ \eta_{C_i} \qquad (*)$$

Since $h' \in \mathrm{Mor}_{\mathscr{C}}(\varinjlim C_i, GD)$, after applying the upper right path, we successively get $h' \circ \alpha_i$ and $\varphi_{C_i,D}(h' \circ \alpha_i)$ If let

$$h := \varphi_{\varinjlim C_i, D}$$

then this guarantees the existence of $h : F(\varinjlim C_i) \to D$ but we still want such an h having the 'commuting'

$$\begin{tikzcd}
\mathrm{Mor}_{\mathcal{D}}(FC_i, D) & \mathrm{Mor}_{\mathcal{C}}(C_i, GD) \arrow[l, "\varphi_{C_i,D}"', "\sim"] \\
\mathrm{Mor}_{\mathcal{D}}(F(\varinjlim C_i), D) \arrow[u, "(-)\circ F\alpha_i"] & \mathrm{Mor}_{\mathcal{C}}(\varinjlim C_i, GD) \arrow[l, "\varphi_{\varinjlim C_i,D}"', "\sim"] \arrow[u, "(-)\circ\alpha_i"']
\end{tikzcd}$$

Figure 6

property. By commutativity of the above diagram,

$$\boxed{h \circ F\alpha_i = \varphi_{C_i,D}(h' \circ \alpha_i)} \qquad ((1))$$

$$\begin{tikzcd}
\mathrm{Mor}_{\mathcal{C}}(C_i, GD') \arrow[r, leftrightarrow, "\varphi_{C_i,D'}", "\sim"'] \arrow[d, "Gf\circ(-)"'] & \mathrm{Mor}_{\mathcal{D}}(FC_i, D') \arrow[d, "f\circ(-)"] \\
\mathrm{Mor}_{\mathcal{C}}(C_i, GD) \arrow[r, leftrightarrow, "\varphi_{C_i,D}", "\sim"'] & \mathrm{Mor}_{\mathcal{D}}(FC_i, D)
\end{tikzcd}$$

Figure 7

Here let $D' = FC_i$, we immediately have the following diagram:

$$\begin{tikzcd}
\mathrm{Mor}_{\mathcal{C}}(C_i, GFC_i) \arrow[r, leftrightarrow, "\varphi_{C_i,FC_i}", "\sim"'] \arrow[d, "G\gamma_i\circ(-)"'] & \mathrm{Mor}_{\mathcal{D}}(FC_i, FC_i) \arrow[d, "\gamma_i\circ(-)"] \\
\mathrm{Mor}_{\mathcal{C}}(C_i, GD) \arrow[r, leftrightarrow, "\varphi_{C_i,D}", "\sim"'] & \mathrm{Mor}_{\mathcal{D}}(FC_i, D)
\end{tikzcd}$$

Figure 8

So, start from $\eta_{C_i} \in \mathrm{Mor}_{\mathscr{C}}(C_i, GFC_i)$, via the upper right path, we have $\gamma_i \circ ((\varphi_{C_i,FC_i})(\eta_{C_i}))$ here φ_{C_i,FC_i} is defined to be the morphism $\varphi_{C_i,FC_i} : \mathrm{Mor}_{\mathscr{C}}(C_i, GFC_i) \to \mathrm{Mor}_{\mathscr{D}}(FC_i, FC_i)$ and η_{C_i} is by definition what def the inverse of φ_{C_i,FC_i}. So,

$$\gamma_i \circ ((\varphi_{C_i,FC_i})(\eta_{C_i})) = \gamma_i$$

Via the lower left path, and by commutativity of this diagram, we have

$$\boxed{\varphi_{C_i,D}(G\gamma_i \circ \eta_{C_i}) = \gamma_i} \qquad (2)$$

$$h \circ F\alpha_i \overset{(1)}{=} \varphi_{C_i,D}(h' \circ \alpha_i) \overset{(*)}{=} \varphi_{C_i,D}(G\gamma_i \circ \eta_{C_i}) \overset{(2)}{=} \gamma_i$$

(1): From the first diagram representing adjoint property
(2): From the second diagram representing adjoint property
($*$): Commutativity

$\square$

4 Internalization

4.1 Some motivations

In topological/Lie/algebraic group theory, we might see the definition is that G is a topological/Lie/algebraic group $\Leftrightarrow$ G is a group and a topological space/smooth manifold/algebraic variety that the following operations

$$m : G \times G \to G \quad \mathrm{inv} : G \to G$$

are *compatible* with the topology/smooth structures/algebraic structures. Namely, the maps m and inv are continuous/smooth/regular.
We can always see that the compatibility means the group operations are the morphisms in each category. But, few of those places formalizes the word 'compatible'.
Take a topological group G as an example. An operation of taking open sets: $G \to \tau(G)$ with $p \mapsto U(\ni p)$ is not a mapping at all(not well-defined). Instead, we lift the context to a set-level. Namely, the multiplication map $m : G \times G \to G$ induces a set-theoretic function

$$m^{-1} : \mathcal{P}(G) \to \mathcal{P}(G \times G) \qquad U \mapsto m^{-1}(U)$$

which is the set-level 'representatives' of the multiplication map m.

Topolgy is **compatible** with the multiplication can be formalized as existence of a mapping $m^* : \tau(G) \to \tau(G \times G)$ that makes the following diagram commute:

$$\begin{tikzcd}
\tau(G \times G) \arrow[d, hook, "\iota_{G\times G}"'] & \tau(G) \arrow[l, "\exists\, m^*"'] \arrow[d, hook, "\iota_G"] \\
\mathcal{P}(G \times G) & \mathcal{P}(G) \arrow[l, "m^{-1}"']
\end{tikzcd}$$

Then, the commutativity implies that for each open set U of G, $m^*(U) = m^{-1}(U)$, meaning the inverse image $m^{-1}(U)$ lies in the topology of $G \times G$. Hence, if m is compatible with the topology of G, m is continuous.

Similarly, the inverse map is said to be **compatible** with the topology of G when $\exists$ a mapping $\mathrm{inv}^* : \tau(G) \to \tau(G)$ that makes the following diagram commute:

$$\begin{tikzcd}
\tau(G) \arrow[d, hook, "\iota_{G\times G}"'] & \tau(G) \arrow[l, "\exists\, \mathrm{inv}^*"'] \arrow[d, hook, "\iota_G"] \\
\mathcal{P}(G) & \mathcal{P}(G) \arrow[l, "\mathrm{inv}^{-1}"']
\end{tikzcd}$$

It follows that inv is continuous from this diagram.

4.2 Internalized objects

4.2.1 Internalized group objects

In the classic BourBaki structural point of view, a topological/Lie/algebraic group is a set equipped with two structures, a topology and an operation. These two structures are parallel and have communication with each other. The 'communication' is called compatibility in the defintion. We just formalized the definition of compatibility. The way we dealt compatibility is very set-theoretic and 'absolute'. This leads us to somewhere:

(1) In category theory, the 'internal' of an object is always a blackbox. We cannot say what are the subsets of an object.

(2) The category theory and Yoneda lemma inspires us to have a relative point of view. We study an object by how it links with any other objects, morphisms between this object and any other ones.
To fit this, one need characterizations of groups in the form of morphisms.
Then, we can define a group object in category $\mathscr{C}$:

Definition 4.1: Group object of category $\mathscr{C}$

Let $\mathscr{C}$ be a category with finite products. A **group object** in $\mathscr{C}$ is an object $G \in \mathrm{Ob}(\mathscr{C})$ together with the following morphisms:
- (multiplication) $G \times G \to G$
- (inverse) $G \to G$
- (identity) $\{*\} \to G$.

The next question is on the study of group objects: all group objects form a category, denoted $\mathbf{Gp}\mathscr{C}$.

Definition 4.2: Group object category

Still, let $\mathscr{C}$ be a category with finite products.

Example (1) When $\mathscr{C} = \mathbf{Top}$, the corresponding group objects are in $\mathbf{Gp}\,\mathbf{Top}$. Objects in $\mathbf{Gp}\,\mathbf{Top}$ are topological groups.
(2) When $\mathscr{C} = \mathbf{Diff}$, the corrspoding group objects are in $\mathbf{Gp}\,\mathbf{Diff}$. Objects in $\mathbf{Gp}\,\mathbf{Diff}$ are Lie groups.

Topological groups can be viewed as group objects in **Top**. Is it possible to reverse the order of viewing? More precisely, is it possible to view topological groups as 'topological space objects' in **Gp**?
At least in this category theory, the answer is no. Because to define topological space, we use subsets when defining open sets. There is no such 'internal view'. So, the first step is blocked.

The category of group objects in **Top** is equivalent to the category of topological groups **TopGp**:

$$\mathbf{Gp}\,\mathbf{Top} \cong \mathbf{TopGp}$$

5 Homotopy theory

5.1 Nerve